Yehya Farhat

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Research Interests: Efficient Training of Large AI Models | Convex & Non-Convex Optimization | Agentic AI

Education

Rice University *Aug 2024 - May 2028*
PhD in Computer Science Houston, TX

- *GPA: 3.9/4.0*
- *Advisor: Prof. Anastasios Kyrillidis*

Syracuse University *Aug 2021 - May 2023*
Master of Science in Computer Science Syracuse, NY

American University of Beirut *Aug 2016 - Dec 2020*
Bachelor of Science in Computer Science, Minor in Mathematics Beirut, LB

Publications & Preprints

Farhat Y., Desmond M., Kyrillidis A.
"Decoupling Memory from Context: Structured Memory for Token-Efficient Test-Time Continual Learning"
Neurips 2026 workshop on Test-Time Continual Learning Agents (TTCL)

Menezes M., Su B., Feng X., **Farhat Y.**, Shili H., Kyrillidis A.
"TwIST: Rigging The Lottery in Transformers With Independent Subnetwork Training"
(Under Review)

Farhat Y.*, Shili H.*, Liao F.*, Dun C.*, Garcia M.*, Zheng G., Awadallah A., Sim R., Dimitriadis D., Kyrillidis A.
"Learning to Specialize: Joint Gating-Expert Training for Adaptive MoEs in Decentralized Settings"
Conference on Neural Information Processing Systems; Neurips-2025

Farhat Y.
"On the Robustness of Decision-Focused Learning"

Farhat Y.
"End-to-End Decision-Focused Learning Using Learned Solvers"

Work Experience

AI Research Intern *May 2026 – Aug 2026*
IBM Research Yorktown Heights, NY

- Developing novel test-time learning techniques to improve the adaptability of agentic systems.
- Designed a persistent-memory architecture that reduced agent token usage by up to 85% relative to state-of-the-art methods while preserving performance.

Data Scientist Intern *June 2022 – August 2022*
Qnovo Milpitas, CA

- Designed and built a battery testing platform with a corresponding user interface to query a database and analyze experimental battery data, reduced time spent on ad-hoc reporting and analysis from 2-3 days to 10 minutes
- Cleaned and performed exploratory data analysis on 5M + rows of time-series data. Analysis was used to confirm the existence of a sensitivity problem in battery capacity estimates
- Implemented predictive models for battery degradation using time-series data, improving data-driven inference pipelines.

Software Engineer *April 2021 – July 2021*
Precast FZCO Dubai, UAE

- Devised a cloud-based micro-architecture and employed Microsoft Azure to establish a correspondence between CMMS software and the client's IoT software, thereby streamlining task creation between software systems and reducing task creation time to zero
- Designed and implemented an integration module to connect multiple ERP, IoT, and CMMS software systems. The module included an interface allowing clients to specify what information to streamline between software systems

Research Experience

Research Assistant *August 2024 – Present*
Rice University
Advisor: Prof. Anastasios Kyrillidis

- Developing scalable and cost-efficient methods for training and deploying large AI system. Research themes: optimization algorithms, distributed training, model compression (pruning/quantization), Mixture-of-Experts, agentic AI & routing, and continual learning.

Research Scholar *September 2023 – June 2024*
Syracuse University
Advisor: Prof. Venkata Gandikota

- Studied the expressiveness of piecewise linear functions attained by deep neural networks through the lens of linear regions. Work involves analyzing the number of regions attained by the learned network and its ability to adapt to different domains based on its expressivity and linear regions.
- Studied extensions for a distributed Quasi-Newton method for general optimization algorithms

Research Assistant (Master's Thesis) *August 2022 – May 2023*
Syracuse University
Advisor: Prof. Ferdinando Fioretto

- Conducted a survey of End-to-End constrained optimization learning and researched an amortized method for the Decision-Focused Learning paradigm. Our framework involves developing a surrogate machine learning model for a continuous relaxation of the discrete optimization problem and substituting the learned model with the optimization solver within the pipeline.
- *"End-to-End Decision-Focused Learning Using Learned Solvers"*
(2023). Theses - ALL. 714. https://surface.syr.edu/thesis/714

Projects

Adapting Transformers for Domain-Specific Reasoning and Higher-Order Logic *May 2022*

- Collaborated with a small team to implement attention-based mechanisms for learning higher-order logic theorem proving. We implemented multiple language models and trained them on the HolStep dataset in an attempt to study automatic theorem proving; *Project Link*

Optimal Stopping Problem with Unknown Variable *December 2021*

- Analyzed one form of the optimal stopping problem known as the secretary problem which has a closed form solution. The equation requires the number of candidates to be known beforehand. Collaborated with a team member to attempt to find an efficient solution to the same problem but without access the number of candidates; *Project Link*

University Petition System *December 2020*

- Collaborated with three team members to develop a course petition website for the American University of Beirut. This platform enables students to submit their petitions along with their transcripts, which are automatically parsed. The chairperson is notified if the students meet the petition requirements.
- This project enabled us to significantly streamline the process, saving the chairperson countless hours previously spent organizing and reviewing transcripts to determine if the requirements have been met;
- Technology stack: JavaScript | Node.js | React | MongoDB
Project Link

Power Monitor *July 2020*

- Cofounded & developed power monitor, pitched to Lebanon's public government energy sector. Enable the sector to transition into digital bills and cut down on the required transportation. Saving the government thousands of dollars each month
- Allows users to track daily/monthly usage, electricity bill, and carbon footprint using the phone application; *Product Website*

Technical Skills

Languages: Python, C/C++, JavaScript, Julia, Dart

ML/DS: PyTorch, Numpy, Pandas, Keras, HuggingFace, LangChain, SciPy, CvxPy, Scikit-Learn

Other: Flask, React, Flutter, SQL, GraphQL, REST API, Selenium, CUDA, HTCondor, Azure, AWS

Fellowship & Awards

NeurIPS Financial Assistance Award 2025
Financial award from the NeurIPS Conference to support registration and accommodation for attendance at the Thirty-Ninth Annual Conference ($\sim 2500\$$).

Engineering Graduate Student Travel Grant 2025
Travel award from Rice Engineering Alumni to support engineering graduate students who are attending conferences or professional meetings to present their research work ($500\$$).

Computer Science Graduate Student Support Grant 2024
Financial award from the Rice University Computer Science Department granted to first year graduate students to support conference participation, Rice-related travel, and research related expenses ($2500\$$).

Teaching Experience

Computational Optimization for AI *Spring 2026*
Teaching Assistant Houston, TX
COMP 282 | Rice University

- Undergraduate Course: Mathematical and Computational Foundations of AI. Taught by *Prof. Anastasios Kyrillidis*

Graduate Design and Analysis of Algorithms *Spring 2025*
Teaching Assistant Houston, TX
COMP 582 | Rice University

- Graduate course: Advanced Design and Analysis of algorithms. Taught by *Prof. Xiaoyun Fu*

Introduction to Object-Oriented Design *Spring 2023*
Teaching Assistant Syracuse, NY
CSE 283 | Syracuse University

- Undergraduate course: Fundamental software design concepts of functional decomposition and object-oriented design. Taught by *Prof. Farzana Rahman*

Computer Organization & Programming Systems *Spring 2023*
Teaching Assistant Syracuse, NY
CIS 341 | Syracuse University

- Undergraduate course: Essentials of computer organization, digital logic, microprogramming, processors, memories, input-output devices. Instruction sets, assembly language, data types, relation to high level languages. Taught by *Prof. Farzana Rahman*

Assured Programming and Formal Methods *Fall 2022*
Teaching Assistant Syracuse, NY
CIS 623 | Syracuse University

- Graduate-level core course: Reasoning about programs through the specification, design, and realization of provably correct programs. Taught by *Prof. Andrew Lee*

Outreach & Extracurricular

President and Co-Founder *February 2022 – October 2022*
Syracuse University Graduate Chess Club Syracuse, NY

- Co-founded the Graduate Chess Club at Syracuse University and served as the Acting President, orchestrating the club's growth with over 150 participants. Merged the meetings of the graduate and undergraduate chess clubs, fostering a cohesive and inclusive chess community.

Blog Posts

- ***Do Neural Networks Have Closed-Form Solutions?***
  Farhat, Y. (2025, December). Do Neural Networks Have Closed-Form Solutions?
- ***How to Build Your Own Custom Optimizer in PyTorch (Step-by-Step)***
  Farhat, Y. (2025, June). How to Build Your Own Custom Optimizer in PyTorch (Step-by-Step). Medium.
- ***What Are the Machines Learning?***
  Farhat, Y. (2023, September). What Are the Machines Learning?